Name _____ Period _____ Date _____

2-1 Lesson Quiz: Vertex Form of a Quadratic Function

2. The function $h(x) = -\frac{1}{2}(x-4)^2 + 3$ represents the height of a bird y over time x, as it flies past the school. What point on the graph represents the greatest height of the bird above the ground?

Ⓐ $(4, 3)$ Ⓑ $(3, 4)$ Ⓒ $\left(\frac{1}{2}, 4\right)$ Ⓓ $\left(3, \frac{1}{2}\right)$

Explain in one sentence why this point is the greatest height. Use the word *vertex*.

3. What is the equation written in vertex form of a parabola with a vertex of $(9, -1)$ that passes through $(7, 7)$?

Ⓐ $y = 2(x-9)^2 - 1$
Ⓑ $y = 2(x+1)^2 - 9$
Ⓒ $y = 2(x-7)^2 + 7$
Ⓓ $y = 7(x-9)^2 - 1$

Show how you found a.

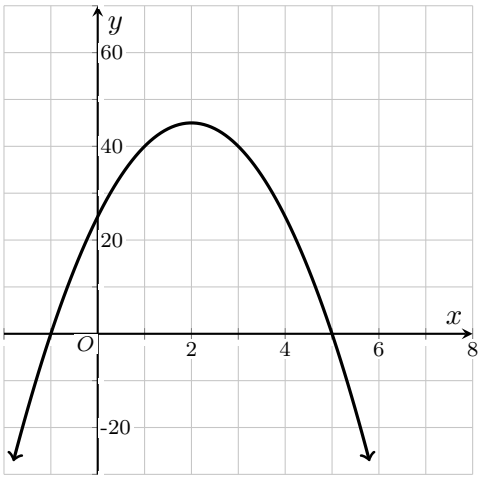
$y = a(x - ____)^2 + ____$

4. Which of the following is an equation in the form $y = ax^2 + bx + c$ of the parabola shown in the graph?

Ⓐ $y = -5x^2 - 20x - 65$
Ⓑ $y = -5x^2 - 20x - 20$
Ⓒ $y = -x^2 - 27x - 90$
Ⓓ $y = -5x^2 + 20x + 25$

From the graph: vertex (_____, _____) y-intercept _____

Show how you found a:



Scoring: Question 2 and Question 4 are 10 points each. Question 3 is a bonus of up to 5 points. Total out of 20. (Questions 1 and 5 of the Savvas quiz are not on this form.)

2-1 Lesson Quiz, short form: answer key

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2. A, $(4, 3)$**.** Matching $-\frac{1}{2}(x-4)^2+3$ to $a(x-h)^2+k$: $a=-\frac{1}{2}$, $h=4$, $k=3$. The vertex is $(4, 3)$; $a<0$, so the parabola opens downward and the vertex is the maximum (greatest height). Distractor B swaps h and k; C and D put the coefficient $\frac{1}{2}$ into the point. DOK 2, skill F60. Taught by Example 2 (Try It 2). **10 points:** 7 for A, 3 for a sentence naming the vertex as the highest point because the parabola opens downward.

3. A, $y=2(x-9)^2-1$**.** $y=a(x-9)^2-1$; substitute $(7, 7)$: $7=a(7-9)^2-1$, $8=4a$, $a=2$. Distractor C puts the point where the vertex goes; D uses the point's y for a. DOK 2, skill F60. Taught by Example 4 steps 1–3 (Example 3 is the same substitution; cut). **Bonus 5:** 5 for A with the substitution shown; 3 for A with no work.

4. D, $y=-5x^2+20x+25$**.** From the graph the vertex is $(2, 45)$ and the y-intercept is 25. $25=a(0-2)^2+45$, $-20=4a$, $a=-5$. $y=-5(x-2)^2+45=-5(x^2-4x+4)+45=-5x^2+20x+25$. Check: $c=25$ is the y-intercept; the zeros $x=-1$ and $x=5$ match the graph. DOK 2, skill F16. Taught by Example 4 (Try It 4). **10 points:** 6 for D, 2 for the vertex and y-intercept read correctly, 2 for $a=-5$ shown.

Items left off this form (available for a retake or Form B): Savvas item 1, select all true statements about $g(x)=(x+2)^2+4$ as a transformation of $f(x)=x^2$ (answer B, C; DOK 2, F60 and F68; Example 1, cut); Savvas item 5, g is $f(x)=x^2$ translated right 3 units and down 5 units, written as $y=ax^2+bx+c$ (answer B, $y=x^2-6x+4$; DOK 2, F16 and F68; Example 5, cut, though its expansion step is Example 4's).

IXL replacement: SmartScore 70 or above on L8C and URV replaces this quiz with 20/20.